8. Beyond the Big Bang

The master

In 1927 a young Belgian scientist, a Jesuit-educated Catholic priest, studies Einstein's equations and realizes – just as Einstein had – that they predict the universe must expand or contract. But instead of foolishly rejecting the result and stubbornly trying to avoid it, as Einstein did, the Belgian priest believes it and looks for astronomical data to test it.

At the time, galaxies were not called galaxies. They were called nebulae because, seen through a telescope, they looked like small, opalescent clouds among the stars. It was not yet known that they are distant, immense islands of stars like our very own galaxy. But the young Belgian priest understands that the scarce available data on the galaxies were indeed compatible with the possibility that the universe is expanding: nearby galaxies are moving away at great speed, as if they had been launched into the sky; distant galaxies are moving away at even greater speed. The universe is swelling like a balloon.

Two years later, the insight is confirmed, thanks to two American astronomers, Henrietta Leavitt and Edwin Hubble. Leavitt discovers a good technique for measuring the distance of the nebulae, confirming that they are very far away, outside of our own galaxy. Using this technique and the great telescope of the Palomar Observatory, Hubble collects precise data that confirm that the galaxies are moving away, at a speed proportional to their distance.